

(20519)

Roll No.

Total Questions : 10]

[Printed Pages : 3

3509

**B.Sc. (Microbiology) IIIrd Year Examination,
May-2019**

INDUSTRIAL MICROBIOLOGY

(B-306)

[B.Sc. (Micro.)]

Time : 3 Hrs.]

[M.M. : 50

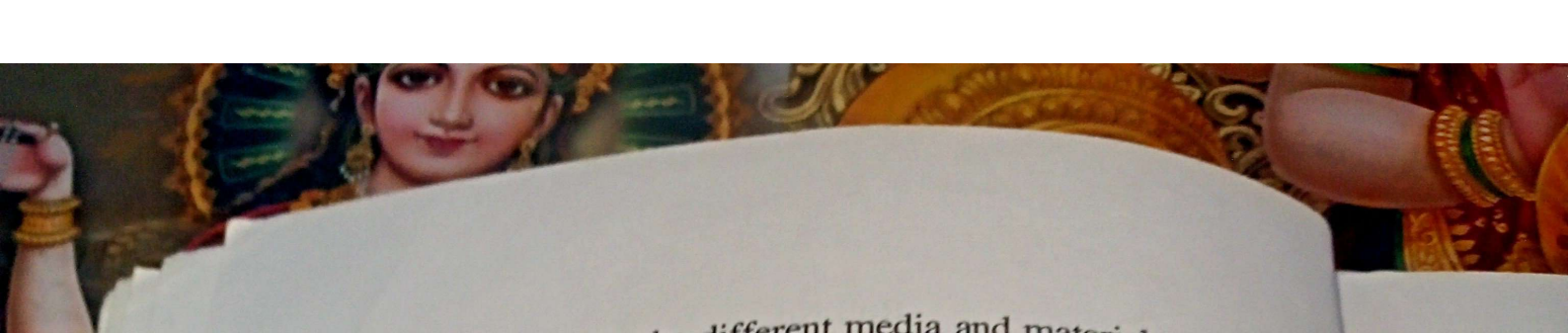
Note :- Attempt *five* questions in all. All questions carry equal marks.

1. Write in detail about the isolation and preservation of Industrially important micro-organisms.
2. Write short notes on the following :
 - (i) Bioleaching
 - (ii) Single Cell Protein (SCP)

NA-340

(1)

Turn Over

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3. What are the different media and materials used in industrial microbiology ?
 4. Write short notes on the following :
 - (i) Mass cultivation of microbes
 - (ii) Steroid transformations
 5. Explain the different types of fermentors and the significance of each fermentor.
 6. Write short notes on the following :
 - (i) Industrial applications of immobilised systems
 - (ii) Explain the production of penicillin
 7. Describe the production of lactic acid from various sources.
 8. How is amylase manufactured ? What are the requirements ?

in

9. Write short notes on the following :

(i) Kinetics of batch culture

(ii) Genetically engineered vaccines

10. Give an account of industrial manufacture of alcoholic beverages.

A
(20620) (Printed Pages 3)
Roll No.
B.Sc. (Micro.) - III Year

3509(CV)

B.Sc. (Microbiology) Examination,

June- 2020

Industrial Microbiology

(B-306)

B.Sc. (Micro.)

Time : Two Hours]

[Maximum Marks : 50

Note : Attempt any **four** questions. **All** questions carry equal marks.

1. What make microbes as most ideal organisms in commercial enterprises? Discuss the use of microbes in industrial products.
2. What do you mean by media? Write down the criteria for media selection and the basic requirements for media formation.

P.T.O.

3. Discuss different methods used for screening of strains. Give the example of industrially important strains discovered for the production of antibiotics.
 4. Draw a neat labelled diagram of submerged fermentor and describe its construction.
 5. Write notes on the following :
 - (i) Industrial applications of immobilised systems.
 - (ii) Microbial acquisition from nature.
 6. Describe applications of physical and chemical sensors in process control in fermentor.
 7. Explain different types of fermentors? Discuss the computer applications in fermentation process.
 8. Give an account of suitable vector and strain of *Bacillus* sp. for large scale fermentation of enzymes & proteins.
- 3509 (CV)/2**

9. Discuss
acids in
fermen
10. Write
(i)
(ii)

9. Discuss the production process of organic acids indicating raw materials used & fermentation method.
10. Write note on the following :
 - (i) Commercial production of Citric acid.
 - (ii) Batch and continuous fermentation.

D
(20721)

(Printed Pages 3)

Roll No. 1.80979131a.21

B.Sc. (Micro.)-III yr.

3509 (CV-II)

B.Sc. (Microbiology) Examination,

June-2021

INDUSTRIAL MICROBIOLOGY

(B-306)

[B.Sc. (Micro.)]

Time : 1½ Hours]

[Maximum Marks : 50

Note : Attempt any **two** questions in all. **All** questions carry equal marks.

1. Discuss the characteristics and maintenance of Industrially important microbes.
2. Write short notes on the following:
 - (i) Significance of media development
 - (ii) What do you mean by Scal-up in Fermentation?

P.T.O.

3. What is Fermentation? Describe the major categories of fermentations and their applications in Industry.
4. Write short notes on the following:
 - (i) Kinetics of Batch culture
 - (ii) Fed Batch fermentation process
5. Describe fermentation technology. Write down their applications in Industry.
6. Write short notes on the following:
 - (i) What factors must be considered in using fermentors?
 - (ii) Importance of Computer in Fermentation.
7. Explain Commercial production of Penicillin and its commercial applications.

8. Describe
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8. Describe in detail the different stages of downstream processing with the help of Flow chart.
9. Write short notes on the following:
 - (i) Solid state fermentation
 - (ii) Product recovery
10. What is the industrial procedure for the production of Lactic acid? Write down its commercial applicators.

A (Printed Pages 3)
(20622) Roll No.
B.Sc.(Micro.)-III Yr.

3509
B.Sc. (Microbiology) Examination,
June-2022

INDUSTRIAL MICROBIOLOGY

(B-306)

[B.Sc. (Micro.)]

Time : Three Hours] [Maximum Marks : 50

Note : Attempt any **five** questions in all. **All** questions carry equal marks.

1. Define fermentation. What parameters can be monitored in a modern large scale industrial fermentation.
2. Describe the industrial procedure for the production of lipase. Write down their commercial applications.

P.T.O.

3. What is the procedure for the production of lysine? Write down their commercial applications.
4. Write short notes on the following:
 - (i) Surface submerged state of fermentation
 - (ii) Significance of media in fermentation
5. Describe Industrial production process of Streptomycin. Write down its commercial applications.
6. Write down the procedure for the production of enzyme amylase. Explain their applications in Industry.
7. Write short notes on the following:
 - (i) Characters of Industrial microbes
 - (ii) Product recovery
8. Describe fermentation technology. Write down their applications in Industry.

9. Write short notes on the following:

(i) Down stream processing in fermentation

(ii) Utility of Computer in fermentation

10. Describe various methods of sterilization.

Why sterilization is important in fermentation.

D
(20524)

B.Sc. (Micro.)-III Year

(Printed Pages 3)

Roll No.

3509

B.Sc. (Microbiology)

Examination, May-2024

INDUSTRIAL MICROBIOLOGY

(B-306)

[B.Sc. (Micro.)]

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. What are the different characters of microorganism responsible for their industrial application? 10
2. Discuss the various habitats of microbes and their relationship with nature. 10

P.T.O.

3. How does different environmental factors affects the microbial growth? 10
4. Discuss the different steps involved in the process of medium development for a microbial process with the help of flow chart. 10
5. Define fermentation. Discuss in detail about submerged and solid state fermentation. 10
6. Explain any **two** in detail. 10
- (i) Batch fermentation
 - (ii) Fed-batch fermentation
 - (iii) Continuous fermentation
7. Give the function of different parts of a fermenter. Show the different parts of fermenter with the help of suitable diagram. 10

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What are the different types of fermenters used in different kind of fermentation processes? Explain the working of any one in detail. 10

9. Give a detailed note on industrial application of microbes for the production of different industrially important metabolites. 10

10. Discuss the production of enzymes in detail with suitable example. 10

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(20425)

(Printed Pages 4)

B.Sc. (Micro.)-III Year

Roll No. 012

3509

B.Sc. (Microbiology)

Examination, April-2025 .

INDUSTRIAL MICROBIOLOGY

(B-306)

[B.Sc. (Micro.)]

Time : Three Hours]

[Maximum Marks : 50

Note : Attempt any **five** questions. **All**
questions carry equal marks.

1. Discuss the key characteristics that make a microbe industrially important. Explain how these characteristics contribute to their industrial applications. 10

P.T.O.

2. (a) What are the different methods used to isolate and identify industrially important microbes from natural environments? Explain in brief. 2×5
- (b) What are the different challenges associated with the isolation of industrially important microbes.
3. Outline the key factors that influence microbial growth in a controlled environment. How these factors can be optimized in a laboratory settings.
4. Discuss following in detail (any **two**): 2×5
- (i) Steps involved in developing a growth media

- (ii) Conditions for optimal microbial growth
- (iii) Scale-up of microbial fermentation
5. (a) Explain microbial growth kinetics in detail. 2×5
- (b) Give a detailed note on different modes of fermentation with suitable example of industrial process.
6. Differentiate single, batch, continuous, surface, submerged and solid state fermentation along with advantages and disadvantages associated with each. 10
7. With the help of well labelled diagram explain the function of different parts of a fermenter/bioreactor. 10

8. Discuss the operation and benefits of stirred-tank, airlift and bubble column fermenters with well labelled diagrams. 10
9. Outline the industrial production process of penicillin. What are the key stages involved in microbial production of penicillin? 10
10. How does industrial amylase production is being carried out? Give a detailed note on the industrial application of amylase enzyme. 10

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(20425)

B.Sc. (M

Time :

Note

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